# Canonical six-bit inverse composition

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**Acceptance.** Synthesize the GF(64) tower inverse, with zero mapped to zero, from own HyMeKo norm, GF(8) inverse and multiplication specifications. Preserve the full arbitrary-target XOR contract, input, phase and clean workspace. Compare separate-output and in-place sub-inverse schedules, and independently check a public arbitrary-target reference in the same tower coordinates. No full DQI replacement, Kaggle upload, optimality or RL claim.

**Ownership and architecture.** Core/Query retain the canonical model. Graph's existing local executable supplies source-bound matroid certificates. Quantum compiles quadratic components, lowers a bounded three-bit involution from its truth function, and composes typed call/binding hyperedges. The existing gate emitter and QuantumModel reader own the flattened circuit. Add an explicit Boolean-permutation execution profile: at most 24 total / 12 data qubits and 512 X/CNOT/Toffoli gates. Reuse one primitive basis-transition implementation; enumerate at most 4096 valid basis states without dense amplitudes. Existing unitary limits stay 9 qubits / 64 gates. Dense probe/execute reject larger widths; probe becomes fallible. Entropy is explicitly unavailable for the sparse profile.

**Changes.** Extend Quantum model/simulation/resource/emitter APIs and quadratic-root selection; add arithmetic composition, metadata, fixture and focused tests. Reuse Python quadratic authoring and introduce a shared native reversible-gate adapter, without copying the existing lowering. Extend the existing Kaggle tool/report workflow. No Graph/Nagare edits or dependency changes. Protected items: none. Preserve unrelated dirty work; no commit or publication.

**Math and schedules.** For x=a+b*y with y*y+y+1=0, compute N=a*a+a*b+b*b, invert N over GF(8), then accumulate (a+b)/N and b/N into the two target halves. Reverse every temporary computation. An in-place GF(8) inverse can be lowered from disjoint transpositions and save three qubits. Compare complete Q-squared-Toffoli proxy costs; nonlinear workspace costs are not summed from individual calls. Components and their bindings remain in Core.

**Verification.** Check all 4096 input/target states, including zero, phase and scratch cleanup, using independent native finite-field arithmetic. Test wrong/foreign witnesses, malformed bindings, unsupported large-profile operators, dirty cleanup, dense allocation refusal and original profile limits. The sparse verifier's exact permutation comparison covers the clean input subspace; no dense-state entropy is fabricated. Existing phase/norm tests remain.

**Budget, provenance and ETA.** Freshly fetched Transformation at 32c6bb3, current with upstream. One CPU, 2 GiB hard / 1800 MiB sampled memory, 900 seconds per explicit native check on Amaterasu; no WSL. Existing Graph binary is an explicit local integration aid, with hash and checked coefficients. No clean-clone claim. ETA about 60–90 minutes including focused checks, report and archive. Final verification uses serialized artifacts. Report once, archive once after writers stop; retain the earlier 883-qubit incumbent.

**Risks.** Classical permutations require a separate guarded execution profile, not a larger unconstrained state vector. A better workspace schedule need not beat a measurement-assisted reference in Toffoli count. All comparisons must use identical input/output/zero contracts. Roll back only named task edits.

**Measured-cleanup continuation, 21:25 CET.** The unitary own inverse passes all 4096 basis states at Q15/T30; public iso20 passes at Q18/T20. Add an explicitly authored inverse-norm erasure macro. Certify the selected unitary prefix on all 4096 states, including its exact remaining scratch value. Measure the inverse norm, recompute the norm, repay the inverse polynomial's classical-mask phase, then measure the norm and repay its quadratic phase from the original inputs. Check all 64 input words and 8-by-8 measurement masks. The native schema becomes arithmetic.v2 for this optional lowering; the retained unitary reference remains in Core. Classical phase corrections use only degree one/two functions, so the expected cleanup costs 6 instead of 9 Toffolis. The native instruction cap is 1024, with the original circuit/width bounds unchanged. Native measurement verification uses three seeds. No speculative replacement is accepted before both the symbolic and native gates pass.

**Outcome.** The measured variant passed all 4096 input/target states at three measurement seeds, with zero value, phase, round-trip and workspace errors. Q15/T27 gives proxy 6075 versus public Q18/T20 at 6480 and Q21/T19 at 8379. The existing native selector now also accepts the generic `reversible_kernel` scope under the same strict microkernel contract checks. 62 targeted Rust and 6 Python tests pass. The native computation moved to isolated /dev/shm after /home filled; outputs were copied and hashed locally. No data was deleted to make space, and no complete decoder was replaced.

